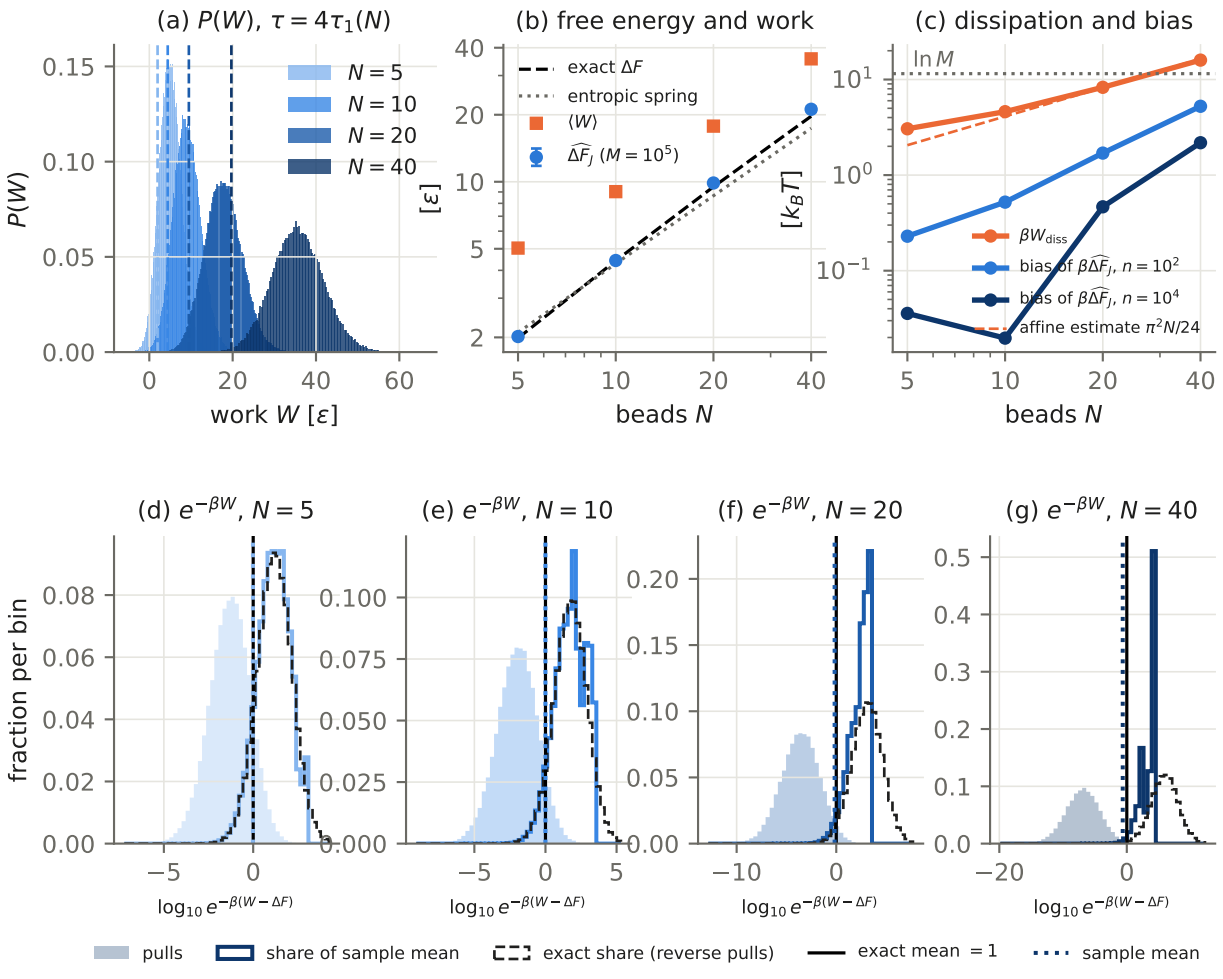
(a) $P(W)$, $\tau = 4\tau_1(N)$
$P(W)$
work W [ε]
$N = 5$
$N = 10$
$N = 20$
$N = 40$
(b) free energy and work
exact ΔF
entropic spring
$\langle W \rangle$
$\widehat{\Delta F}_j$ ($M = 10^5$)
[ε]
[$k_B T$]
beads N
(c) dissipation and bias
$\ln M$
βW_{diss}
bias of $\beta\widehat{\Delta F}_j$, $n = 10^2$
bias of $\beta\widehat{\Delta F}_j$, $n = 10^4$
affine estimate $\pi^2 N/24$
beads N
(d) $e^{-\beta W}$, $N = 5$
(e) $e^{-\beta W}$, $N = 10$
(f) $e^{-\beta W}$, $N = 20$
(g) $e^{-\beta W}$, $N = 40$
fraction per bin
$\log_{10} e^{-\beta(W-\Delta F)}$
pulls
share of sample mean
exact share (reverse pulls)
exact mean = 1
sample mean